Angles & 2D Shapes

Year 4 · **Lesson Worksheet**

Worked Examples

Easy: What type of angle is 40°?
Answer: It is between 0° and 90°, so it is acute.

Standard: What type of angle is 130°?
Answer: It is between 90° and 180°, so it is obtuse.

Advanced: Two angles on a straight line are 110° and x°. What is x?
Answer: Angles on a straight line add to 180°. x = 180 − 110 = 70°.

Easy / Standard (15 questions)

1. What type of angle is 30°?
2. What type of angle is 100°?
3. Which of these is a right angle? A) 45° B) 90° C) 135° D) 180°
4. What type of angle is exactly 90°?
5. How many sides does a triangle have?
6. How many sides does a square have?
7. Which shape has 4 equal sides and 4 right angles? A) rectangle B) square C) triangle D) pentagon
8. How many sides does a pentagon have?
9. Select ALL of these angles that are ACUTE (less than 90°): A) 45° B) 90° C) 20° D) 100° E) 75°
10. How many sides does a hexagon have?
11. What is the name for an angle that is exactly 180°?
12. Which shape has 3 sides? A) square B) triangle C) pentagon D) hexagon
13. How many lines of symmetry does a square have?
14. What type of angle is 170°?
15. How many sides does a rectangle have?

Standard (15 questions)

1. What type of triangle has all three sides equal?
2. Which triangle has exactly two sides equal? A) scalene B) isosceles C) equilateral D) right-angled
3. What type of triangle has no equal sides?
4. How many lines of symmetry does a rectangle (non-square) have?

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5. Angles on a straight line always add up to how many degrees?

6. Which of these describes a right- angled triangle? A) it has one angle of exactly 90° B) all its angles are equal C) it has no equal angles D) it has one angle over 90°

7. Angles around a point always add up to how many degrees?

8. Select ALL of the following that are quadrilaterals (4-sided shapes): A) square B) rectangle C) triangle D) pentagon E) rhombus

9. How many lines of symmetry does an equilateral triangle have?

10. What is the sum of the interior angles of a triangle?

11. Which of these shapes has NO straight sides? A) circle B) square C) triangle D) hexagon

12. A right angle is split into two smaller angles. If one is 30°, what is the other?

13. How many sides does an octagon have?

14. Which shape is a rhombus most like? A) a square with slanted sides B) a circle C) a triangle D) a pentagon

15. How many right angles does a rectangle have?

Standard / Hard (15 questions)

1. On a straight line sit two angles, one measuring 55° and the other labelled x°. Work out the value of x.

2. Three angles meet at a single point, measuring 100°, 130° and x°. What value must x take?

3. If two angles of a triangle are 40° and 80°, use the fact that all three interior angles always add to 180° to find the missing third one. A) 40° B) 60° C) 70° D) 100°

4. Where two straight lines cross, they create pairs of vertically opposite angles. Given that one of these angles is 55°, what size is the angle directly opposite it?

5. In an isosceles triangle, two of the angles are equal, each measuring 65°. What must the third angle be?

6. Three of a quadrilateral's angles are 90°, 90° and 100°. What size is its fourth angle? A) 70° B) 80° C) 90° D) 100°

7. Some facts about shapes are listed below. Select ALL of the ones that are actually TRUE: A) A square counts as a type of rectangle B) A rectangle always counts as a square C) Every rhombus has 4 sides of equal length D) Every triangle has 3 sides of equal length E) A square always has 4 lines of symmetry

8. A right angle gets divided into two smaller angles. One part measures 35° — what size is the other part?

9. A straight line is divided into two angles that happen to be equal to one another. What size must each one be?

10. Out of the four shapes listed, which one has just a single line of symmetry? A) a square B) a regular hexagon C) an equilateral triangle D) an isosceles triangle, but one that is not equilateral

11. A triangle contains a 90° angle, and its remaining two angles are equal to each other. What size is each of those two angles?

12. An obtuse-angled triangle always has one angle greater than 90°. Select ALL of the angle sets below that could belong to such a triangle: A) 100°, 40°, 40° B) 60°, 60°, 60° C) 120°, 35°, 25° D) 90°, 45°, 45° E) 95°, 55°, 30°

13. A quadrilateral's four interior angles always add up to 360°. Given that three of them measure 110°, 90° and 70°, what must the fourth one be?

14. The interior angles of any pentagon add up to 540° in total. If four of this pentagon's angles are each exactly 100°, what size is the fifth?

15. A straight line carries two angles of 70° and 110°. Are these two angles the same size as each other, and what total do they make together?

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